



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO NISHALL FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

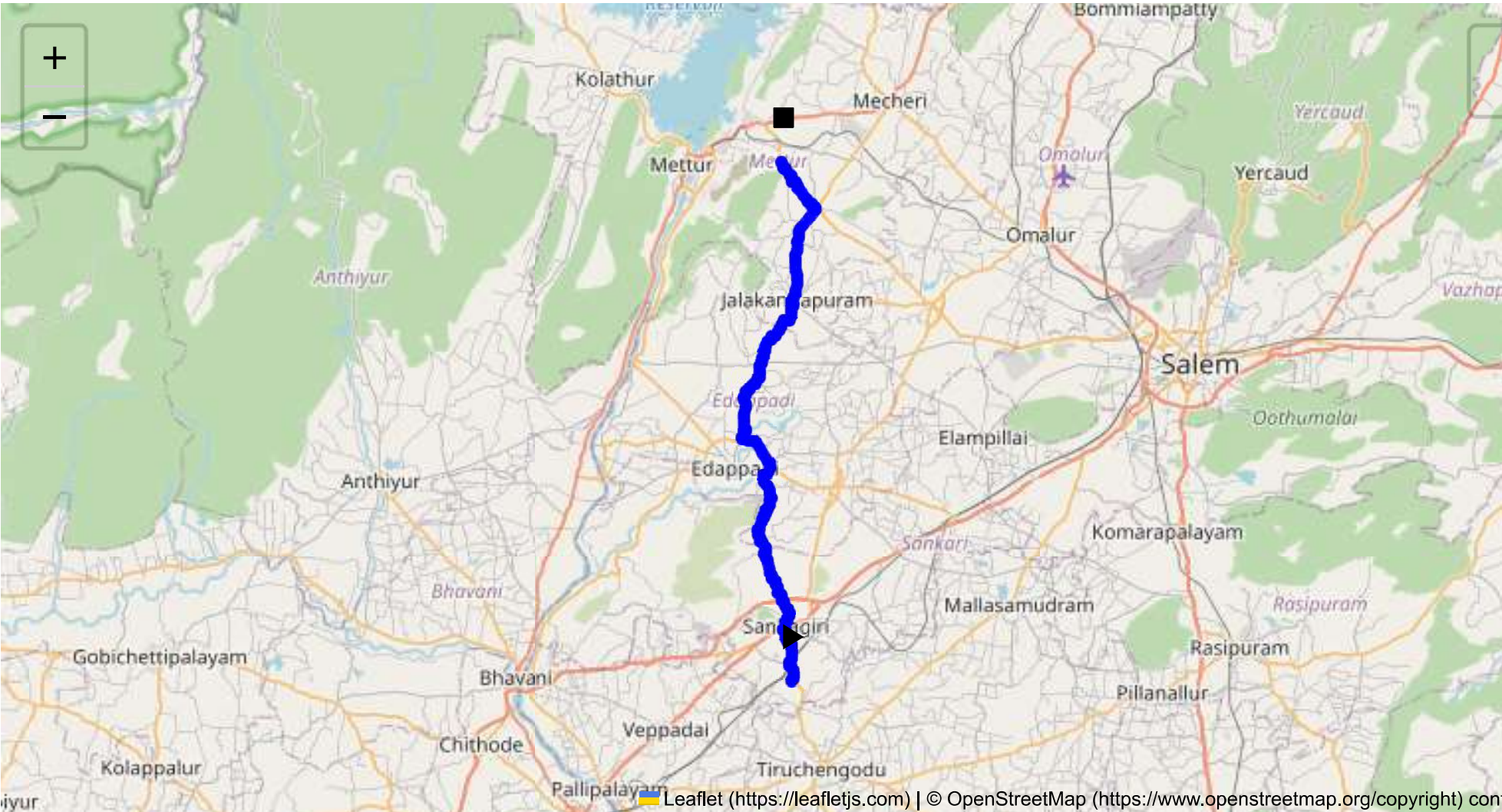
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 45.93 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.792024, 77.86654)



Welcome to the Journey Risk Management Study

Route Safety Analysis for Route: Sangagiri to Mangamethai, Veerakkal, Tamil Nadu

1. Overview of the Route Map

The route from Sangagiri to Mangamethai spans approximately 45.93 kilometers. Starting in Sangagiri, you would likely travel along the NH79, a major road, until close to Salem and then transition to smaller roads leading to Mangamethai. The route navigates through both urban areas and sections of rural Tamil Nadu, highlighting a mix of highway and local roads.

2. Typical Weather Conditions and Potential Hazards

- **Weather:** This region experiences a tropical climate. Summers can be extremely hot (up to 40°C), and the monsoon season (June to September) brings heavy rains. Winters are mild and pleasant.
- **Weather Hazards:** During the monsoon, heavy rains can lead to waterlogging and reduced visibility. High temperatures in summer could lead to vehicular overheating.

3. Traffic Patterns

- **Peak Hours:** Expect traffic to be heavier during morning and evening rush hours, approximately 8-10 AM and 5-7 PM.
- **Congestion-Prone Areas:** Near the Salem urban area and in small market towns along the route where roads can become congested due to local activity.

4. Road Quality and Infrastructure

- **NH79 is generally in good condition**, with marked lanes and sufficient signage.
- Smaller local roads towards the destination may have poorer surfaces and less reliable markings.
- **Infrastructure Issues:** Watch for narrow bridges and turns in rural and mountainous sections.

5. Suggestions for Alternative Routes

- In case of severe congestion or roadblocks, consider taking NH44 which runs parallel and intersects with other major roads leading to the destination. Always have an updated GPS/device to find local road diversions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits and Restrictions:** Ensure you carry valid transport permits. Movement of hazardous materials may be restricted during festivals and public events.
- Compliance with national regulations (such as labels and documentation) is required.

7. Overview of Historical Incidents

- **Accidents:** The region around Salem has reported occasional accidents involving heavy vehicles, typically due to high speeds or overtaking in inadequate conditions.
- **Hazardous Material Spills:** No major historical incidents involving hazardous materials have been widely reported, but caution is advised.

8. Environmental Considerations

- Sensitive areas include minor forest reserves and agricultural zones. Ensure no spillage occurs as it could affect local agriculture.
- Any emissions from the truck should be minimized due to nearby population centers.

9. Analysis of Communication Coverage

- Most of the route, particularly along NH79, enjoys good mobile coverage.
- **Potential Dead Zones:** Remote rural areas and certain road segments in the hills may have intermittent coverage.

10. Estimated Emergency Response Times

- **Urban and Main Highway Areas:** Emergency services (fire, medical) potentially accessible within 15-30 minutes.
- **Rural Areas:** Response times can be significantly longer, often exceeding 45 minutes due to road conditions and distance from response centers.

12. Overall Summary of Risk Assessment

- **Overall Route Safety:** This route has a moderate risk due to traffic congestion, potential weather challenges, and road quality. However, adherence to safety protocols should mitigate most risks.
- **Key Risks Include:** Monsoonal weather hazards, congestion in urban sections, and limited infrastructure on rural roads.
- **Recommended Precautions:** Check weather forecasts, carry essential vehicle maintenance tools, ensure complete documentation for material transport, and communicate anticipated arrival times to ensure timely alerts in case of delays or emergencies.

By understanding these aspects, the truck driver can be prepared for potential obstacles and manage their trip safely and efficiently.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	High	11.45004, 77.87206	15 KM/Hr
7	Turn	Medium	11.45028, 77.87202	30 KM/Hr
8	Turn	Medium	11.45309, 77.87325	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.47451, 77.86914	15 KM/Hr
11	Turn	High	11.47920, 77.86989	15 KM/Hr
12	Turn	Medium	11.48070, 77.87166	30 KM/Hr
13	Turn	Medium	11.48155, 77.87174	30 KM/Hr
14	Turn	Medium	11.48277, 77.87250	30 KM/Hr
15	Turn	Medium	11.49496, 77.86660	30 KM/Hr
16	Turn	Medium	11.52571, 77.85522	30 KM/Hr
17	Turn	Medium	11.53598, 77.85142	30 KM/Hr
18	Turn	Medium	11.53857, 77.85087	30 KM/Hr
19	Turn	Medium	11.55027, 77.85506	30 KM/Hr
20	Turn	Medium	11.55460, 77.85624	30 KM/Hr
21	Turn	High	11.57893, 77.85430	15 KM/Hr
22	Turn	Medium	11.57903, 77.85441	30 KM/Hr
23	Turn	Medium	11.60960, 77.84159	30 KM/Hr
24	Turn	Medium	11.61795, 77.84020	30 KM/Hr
25	Turn	Medium	11.64055, 77.84600	30 KM/Hr
26	Turn	Medium	11.64173, 77.84878	30 KM/Hr
27	Turn	Medium	11.64522, 77.85113	30 KM/Hr
28	Turn	Medium	11.66451, 77.85595	30 KM/Hr
29	Turn	Medium	11.67322, 77.86045	30 KM/Hr
30	Turn	Medium	11.67368, 77.86223	30 KM/Hr
31	Turn	Medium	11.68196, 77.86690	30 KM/Hr
32	Turn	High	11.68488, 77.86889	15 KM/Hr
33	Turn	Medium	11.68463, 77.87262	30 KM/Hr
34	Turn	Medium	11.68486, 77.87288	30 KM/Hr
35	Turn	Medium	11.69289, 77.87444	30 KM/Hr
36	Blind Spot	Blind Spot	11.69469, 77.87589	10 KM/Hr
37	Turn	High	11.69491, 77.87546	15 KM/Hr
38	Blind Spot	Blind Spot	11.69510, 77.87550	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	High	11.69551, 77.87442	15 KM/Hr
40	Turn	Medium	11.69908, 77.87472	30 KM/Hr
41	Turn	Medium	11.69934, 77.87500	30 KM/Hr
42	Turn	Medium	11.76019, 77.89047	30 KM/Hr
43	Blind Spot	Blind Spot	11.76020, 77.89061	10 KM/Hr
44	Turn	High	11.76077, 77.89051	15 KM/Hr
45	Turn	Medium	11.76080, 77.89029	30 KM/Hr
46	Turn	High	11.76245, 77.88739	15 KM/Hr
47	Turn	High	11.76369, 77.88758	15 KM/Hr
48	Turn	Medium	11.76458, 77.88551	30 KM/Hr
49	Turn	Medium	11.78004, 77.87346	30 KM/Hr
50	Turn	Medium	11.78153, 77.87366	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
9	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
10	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
12	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47451, 77.86914



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47920, 77.86989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48070, 77.87166



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48155, 77.87174



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49496, 77.86660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53598, 77.85142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53857, 77.85087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55027, 77.85506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60960, 77.84159



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64055, 77.84600



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64173, 77.84878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64522, 77.85113



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66451, 77.85595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67322, 77.86045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67368, 77.86223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68488, 77.86889



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68486, 77.87288



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69289, 77.87444



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.69469, 77.87589



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69491, 77.87546



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69510, 77.87550



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69551, 77.87442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69908, 77.87472



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69934, 77.87500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.76020, 77.89061



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76077, 77.89051



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76080, 77.89029



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76245, 77.88739



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76369, 77.88758



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76458, 77.88551



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78004, 77.87346



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78153, 77.87366

